Braking judder - guided synthetic demonstration

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1. Executive assessment

Verdict: The measurements support FL disc-thickness variation as a likely source of braking-specific wheel-order excitation; the documented steering response may amplify it. Physical causation remains provisional.

Provisional engineering assessment from the recorded evidence; the physical cause still requires confirmation by inspection.

Selected explanation: Brake disc thickness variation - supported. [E04] [E11] [E13] [E02] [E03]

2. Scope and operating conditions

Operator-planned synthetic demonstration with operator report corrections. Separate DAQ and repeat recordings use source-local clocks; no cross-source alignment was measured or applied. References: BRAKE-SETUP-R1 and BRAKE-QUALITY-R1, both revision R1.

Reference documents

- BRAKE-QUALITY-R1 (revision R1; source SHA256 4ecc4044ca3b253f)
- BRAKE-SETUP-R1 (revision R1; source SHA256 8865cf79fa17562e)

3. Data quality and preparation

The documented WHL_SPD_FL code 6553.5 was replaced with NaN on a derived channel, preserving raw data and gaps. Native output verification found 33 non-finite values among 3200 rows. No undocumented unit or polarity correction was applied. The comparisons use magnitude/RMS rather than correlation sign.

Recording quality summary

Table 1: Recording quality summary

Source	Channels	Rows	Start row	Non-finite	Timestamp gaps	Evidence
daq.mpai	24	409,600	0	0 / 24	0 / 24	[E02]
repeat.mpai	24	409,600	0	0 / 24	0 / 24	[E03]
metrology.csv	5	12	0	0 / 5	0 / 5	[E04]
controller.csv	1	3,200	0	1 / 1	0 / 1	[E06]
controller.csv	800	3,200	0	0 / 800	0 / 800	[E07]
repeat.mpai	24	409,600	0	0 / 24	0 / 24	[E15]
daq.mpai	24	409,600	0	0 / 24	0 / 24	[E16]
metrology.csv	5	12	0	0 / 5	0 / 5	[E17]
controller.csv	800	3,200	0	0 / 800	0 / 800	[E18]

Quality counts show affected / assessed channels. Full channel profiles are available in the connected results and evidence companion. The channel profiles do not assess clipping against declared sensor ranges.

Verified corrected-channel measurements

Table 2: Verified corrected-channel measurements

Channel	Minimum	Maximum	Mean	Valid / total	Evidence
WHL_SPD_FL__find_replace	60.15	120	90.0747	3167 / 3200	[E06]

4. Measured findings

Physical-encoder order analysis covers the complete 0-8 s braking and 8-16 s coast windows. At the same native row of 798.732 rpm (800 rpm requested), steering_accel 1X peak amplitudes are 0.205628 g during braking and 0.00411901 g during coast. These are peak magnitudes, not total RMS. Full-record hub_a/hub_b RMS values are 0.0559551/0.0493958 g in DAQ and 0.0559485/0.0494028 g in the primary-remounted repeat, each over 409600 samples. Calibrated FL thickness spans 27.985-28.015 mm: its native peak-to-peak range of 0.030 mm exceeds the fixture limit of 0.015 mm. Other corners have 0.002 mm ranges.

Measured-encoder order analysis coverage

Table 3: Measured-encoder order analysis coverage

Recording	Physical encoder	Vibration channel	Rows	Evidence
daq.mpai	wheel_encoder	steering_accel	409,600	[E09]

Requested channel measurements

Table 4: Requested channel measurements

Source	Channel	Unit	RMS	Sample std	Valid / total
S02 [E02]	steering_accel	g	0.0709294	0.0709295	409600 / 409600
S02 [E02]	hub_a	g	0.0559551	0.0559552	409600 / 409600
S02 [E02]	hub_b	g	0.0493958	0.0493958	409600 / 409600
S03 [E03]	steering_accel	g	0.0709275	0.0709276	409600 / 409600
S03 [E03]	hub_a	g	0.0559485	0.0559486	409600 / 409600
S03 [E03]	hub_b	g	0.0494028	0.0494028	409600 / 409600
S05 [E06]	WHL_SPD_FL__find_replace		91.7249	17.3232	3167 / 3200
S05 [E07]	WHL_SPD_FL		671.738	653.3	3200 / 3200
S03 [E15]	steering_accel	g	0.0709275	0.0709276	409600 / 409600
S03 [E15]	hub_a	g	0.0559485	0.0559486	409600 / 409600
S03 [E15]	hub_b	g	0.0494028	0.0494028	409600 / 409600
S02 [E16]	steering_accel	g	0.0709294	0.0709295	409600 / 409600
S02 [E16]	hub_a	g	0.0559551	0.0559552	409600 / 409600
S02 [E16]	hub_b	g	0.0493958	0.0493958	409600 / 409600
S05 [E18]	WHL_SPD_FL		671.738	653.3	3200 / 3200

Retained cached scalar rows for explicitly requested channels and derived outputs; other channels remain in the full native review. [E02]: 409600 samples, starting at row 0; [E03]: 409600 samples, starting at row 0; [E06]: 3200 samples, starting at row 0; [E07]: 3200 samples, starting at row 0; [E15]: 409600 samples, starting at row 0; [E16]: 409600

samples, starting at row 0; [E18]: 3200 samples, starting at row 0. Separate-source statistics do not establish time alignment or matching operating conditions. Use the documented setup and intervention before interpreting differences.

Measurement overview - Order Tracking / daq.mpai / steering_accel [E10]

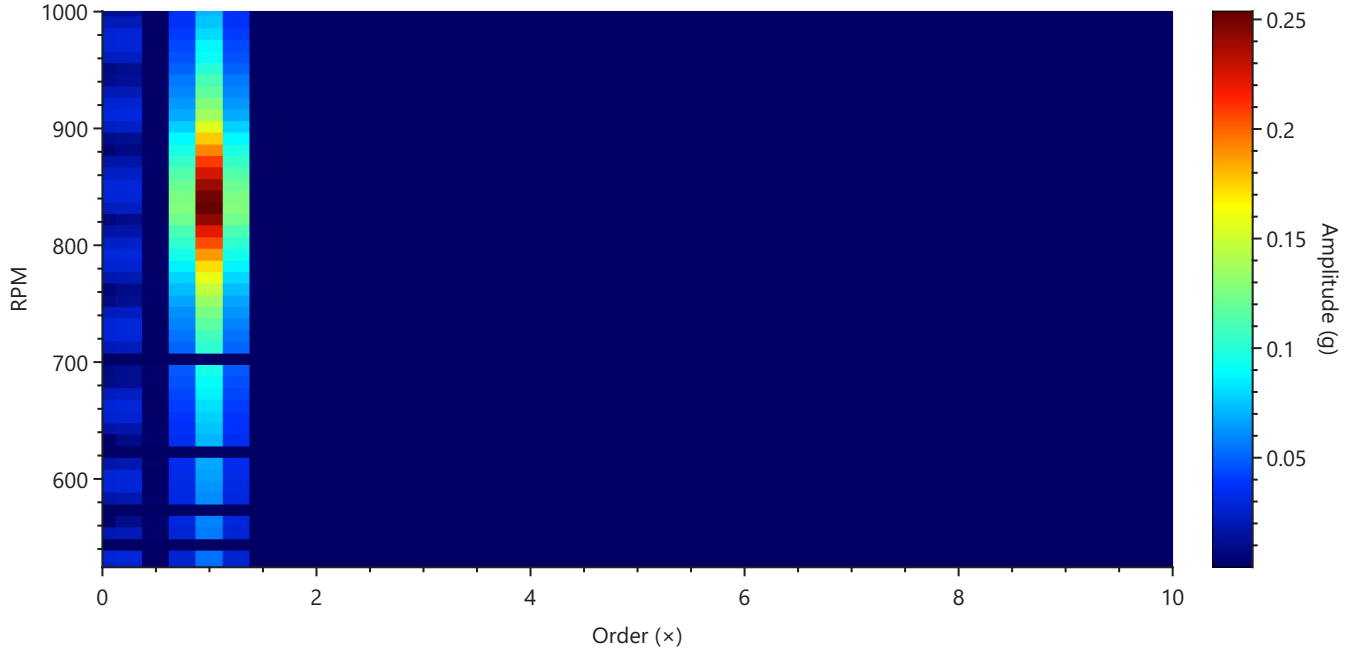


Figure 1: Source: Order Tracking

Processing and operating conditions [E10]

Table 5: Processing and operating conditions [E10]

Parameter	Value
Sample rate (Hz)	6,400
Window	hann

Measurement overview - Order Tracking / daq.mpai / steering_accel [E12]

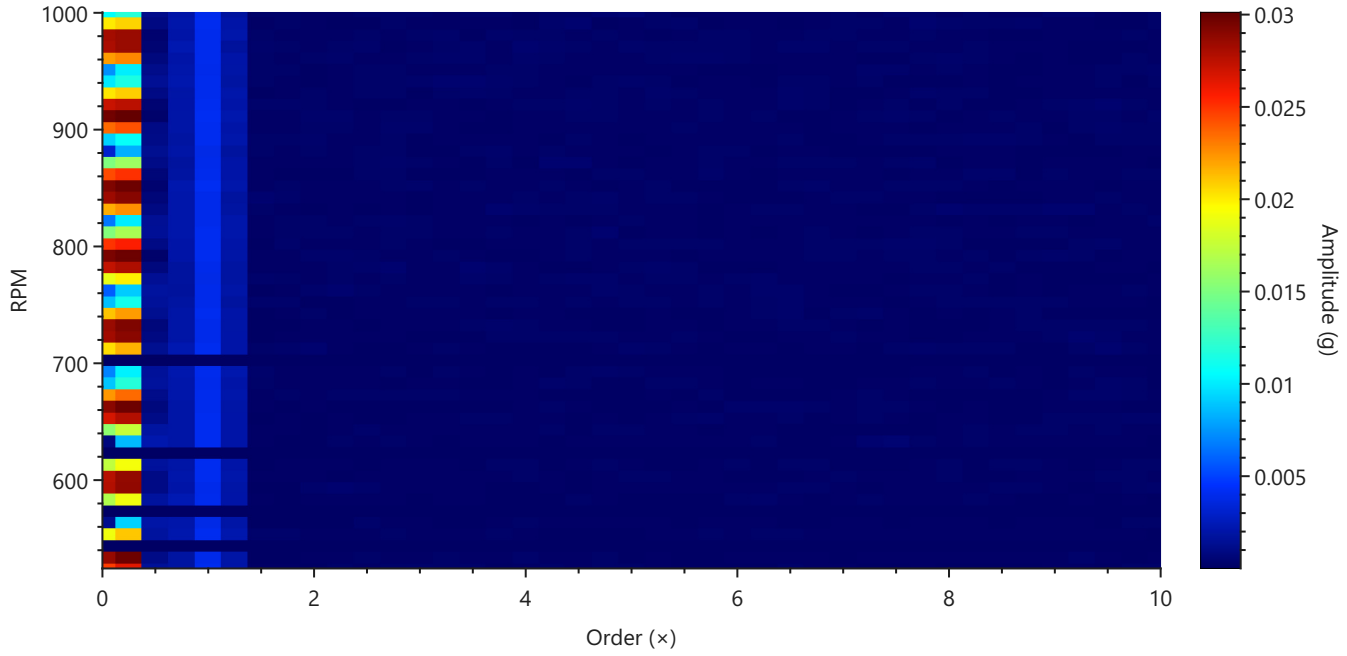


Figure 2: Source: Order Tracking

Processing and operating conditions [E12]

Table 6: Processing and operating conditions [E12]

Parameter	Value
Sample rate (Hz)	6,400
Window	hann

Measurement overview - Order Tracking / daq.mpai / steering_accel [E09]

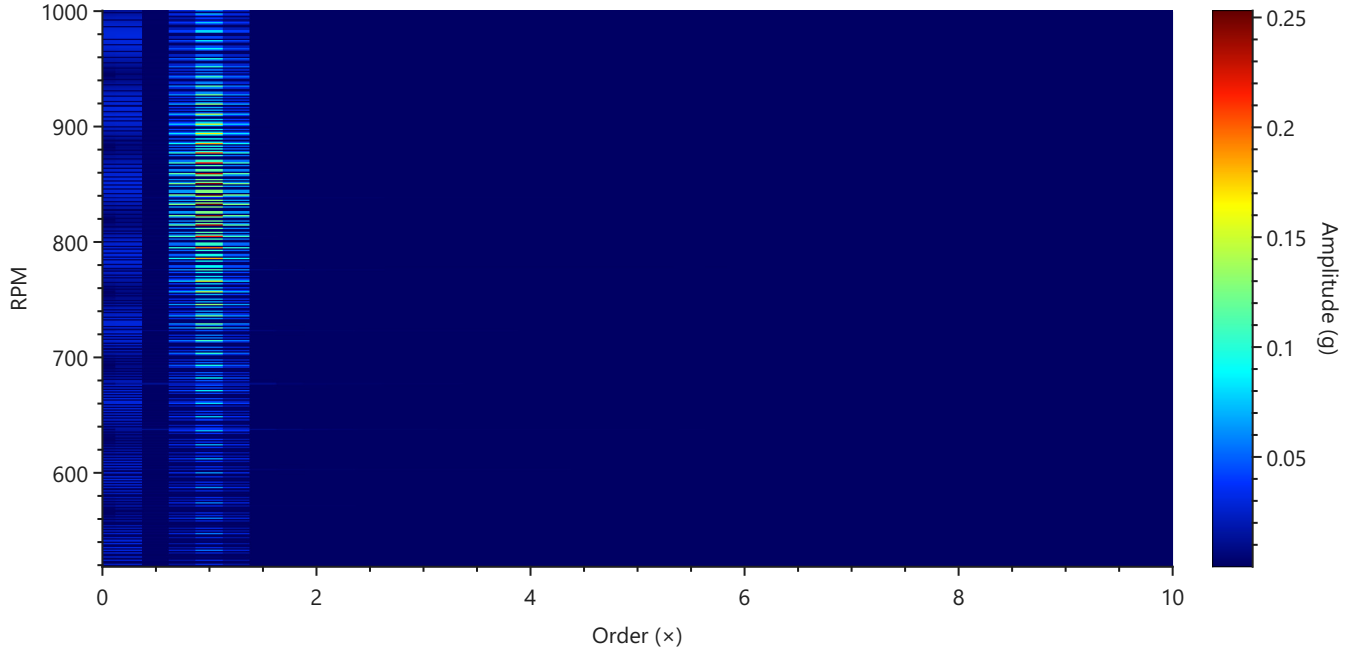


Figure 3: Source: Order Tracking

Processing and operating conditions [E09]

Table 7: Processing and operating conditions [E09]

Parameter	Value
Sample rate (Hz)	6,400
Window	hann

Selected numerical evidence - Channel Profile / metrology [E04] [E17]

Table 8: Selected numerical evidence - Channel Profile / metrology [E04] [E17]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
angle_deg		12	194.808	1.69398	-1.21678	0	330
thickness_FL		12	28	1.00054	-1.5	27.985	28.015
thickness_FR		12	28	1.00004	-1.5	27.999	28.001
thickness_RL		12	28	1.00004	-1.5	27.999	28.001
thickness_RR		12	28	1.00004	-1.5	27.999	28.001

Selected engineering fields; other columns remain available in the connected result.

Selected numerical evidence - Channel Profile / controller [E06]

Table 9: Selected numerical evidence - Channel Profile / controller [E06]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
WHL_SPD_FL__find_replace		3,200	91.7249	1.30826	-1.19999	60.15	120

Selected engineering fields; other columns remain available in the connected result.

Selected numerical evidence - Relationship Explorer / controller [E08]

Table 10: Selected numerical evidence - Relationship Explorer / controller [E08]

Target	Channel	Pearson r	Valid pairs	Status
EPS_TORQUE_RMS	AUX_0129	-0.83906	3,200	Valid
EPS_TORQUE_RMS	AUX_0354	-0.838976	3,200	Valid
EPS_TORQUE_RMS	AUX_0309	-0.838817	3,200	Valid
EPS_TORQUE_RMS	AUX_0759	-0.838777	3,200	Valid
EPS_TORQUE_RMS	AUX_0714	-0.838695	3,200	Valid
EPS_TORQUE_RMS	AUX_0489	-0.838635	3,200	Valid
EPS_TORQUE_RMS	AUX_0084	-0.83861	3,200	Valid
EPS_TORQUE_RMS	AUX_0579	-0.838549	3,200	Valid

Display excerpt: rows 1-8 of 799. All rows remain in the connected result; analysis and acceptance checks use the complete result.

Selected engineering fields; other columns remain available in the connected result.

Relationship interpretation [E08]

Highest returned associations: AUX_0129 r=-0.83906; AUX_0354 r=-0.838976. The coefficients compare paired samples

within the controller source at the reported lag. They do not establish DAQ-to-controller synchronization or a physical cause. Similar coefficients for multiple channels do not constitute independent corroboration; check derived channels and collinearity. Small absolute coefficients provide little linear association in this selection; statistical significance and nonlinear effects are not established here.

5. Engineering interpretation

Interpretation

The combined measurements support FL brake excitation associated with excessive disc-thickness variation. A documented steering response near 14 Hz may amplify the excitation; resonance alone does not identify its source. The persistent independent responses weaken a primary-mounting-only explanation. The much smaller coast response also weakens a wheel-imbalance-only explanation, without excluding residual imbalance.

Sensor mounting problem - rejected

A primary-sensor mounting problem alone explains the braking-specific steering judder.

Supporting evidence: None recorded

Counterevidence: [E02], [E03], [E11], [E13]

Reason: A difference between two sensors alone does not establish mounting damage. Both independent responses persist in the remount-only repeat, and the steering 1X response is much larger during braking than coast at the same measured speed. This rejects the installation-only explanation without excluding every installation issue.

Brake disc thickness variation - supported

The measurements support FL disc-thickness variation as a likely source of braking-specific wheel-order excitation; the documented steering response may amplify it. Physical causation remains provisional.

Supporting evidence: [E04], [E11], [E13], [E02], [E03]

Counterevidence: None recorded

Reason: The calibrated FL thickness range exceeds the documented fixture limit, the same-speed steering 1X amplitude is braking-specific, and both independent sensor responses persist after primary-only remounting.

6. Uncertainty and follow-up

Uncertainty

The association does not prove the complete physical causal chain, a successful repair or operational release. The limit is a synthetic fixture specification, not a maintenance authority.

Recommended checks

Independently inspect the FL disc, hub and brake assembly, then repeat the same measured-speed comparison after intervention.

Required follow-up and limitations

- Independent FL disc/hub inspection
- Matched post-intervention braking/coast repeat

- Supported hypotheses are provisional; physical cause and repair effectiveness require independent verification.
- Rejection concerns the tested explanation in recorded conditions; it does not rule out every physical defect.

7. Technical appendix - evidence register

Evidence references below identify the recorded computations and their source scope.

Report identity

Created: 2026-11-13T07:40:41+00:00
Status: Draft engineering assessment
Author: Application-assisted investigation; analyst/model identity not supplied.
Software: MachinePulseAI; build identifier not supplied

Computation receipts

Table 11: Computation receipts

Evidence	Source	Check	Channels	Status	Selection
[E01]	[S01]	Reference lookup		ok	Not recorded / reference only
[E02]	[S02]	Analysis	Channel list in JSON	Earlier result; latest is [E16]	409,600 samples from sample 0
[E03]	[S03]	Analysis	Channel list in JSON	Earlier result; latest is [E15]	409,600 samples from sample 0
[E04]	[S04]	Analysis	Channel list in JSON	Earlier result; latest is [E17]	12 samples from sample 0
[E05]	[S05]	Analysis		Latest result	Not recorded / reference only
[E06]	[S05]	Analysis	WHL_SPD_FL__find_replace	Latest result	3,200 samples from sample 0
[E07]	[S05]	Analysis	Channel list in JSON	Earlier result; latest is [E18]	3,200 samples from sample 0
[E08]	[S05]	Analysis	Channel list in JSON	Latest result	3,200 samples from sample 0
[E09]	[S02]	Analysis	wheel_rpm, wheel_encoder, steering_accel, ['steering_accel']	Latest result	409,600 samples from sample 0
[E10]	[S02]	Analysis	wheel_rpm, wheel_encoder, steering_accel, ['steering_accel']	Latest result	51,201 samples from sample 0
[E11]	[S02]	Result readback	wheel_rpm, wheel_encoder, steering_accel, ['steering_accel']	ok	51,201 samples from sample 0
[E12]	[S02]	Analysis	wheel_rpm, wheel_encoder, steering_accel, ['steering_accel']	Latest result	51,201 samples from sample 51,200
[E13]	[S02]	Result readback	wheel_rpm,	ok	51,201 samples from

Evidence	Source	Check	Channels	Status	Selection
			wheel_encoder, steering_accel, ['steering_accel']		sample 51,200
[E14]	[S06]	Analysis		Latest result	Not recorded / reference only
[E15]	[S03]	Analysis	Channel list in JSON	Latest result	409,600 samples from sample 0
[E16]	[S02]	Analysis	Channel list in JSON	Latest result	409,600 samples from sample 0
[E17]	[S04]	Analysis	Channel list in JSON	Latest result	12 samples from sample 0
[E18]	[S05]	Analysis	Channel list in JSON	Latest result	3,200 samples from sample 0

Full evidence identities, source paths and processing receipts are retained in the exported .evidence.json companion.

Source register

Table 12: Source register

Source	Recording	Analysed file state
S01	Reference material	Not recorded
S02	daq.mpai	85.18 MB, modified 2026-11-13 10:33:09 +0300
S03	repeat.mpai	85.18 MB, modified 2026-11-13 10:33:09 +0300
S04	metrology.csv	metrology_7867df0ccf44.mpai: 6.53 kB, modified 2026-11-13 10:34:01 +0300
S05	controller.csv	controller_1f81f73d60b3.mpai: 23.71 MB, modified 2026-11-13 10:34:35 +0300
S06	Reference material	Not recorded

Applicable reference register

Table 13: Applicable reference register

Reference	Document	Revision	Evidence
BRAKE-QUALITY-R1	quality.md	R1	[E01]
BRAKE-SETUP-R1	setup.md	R1	[E01]